

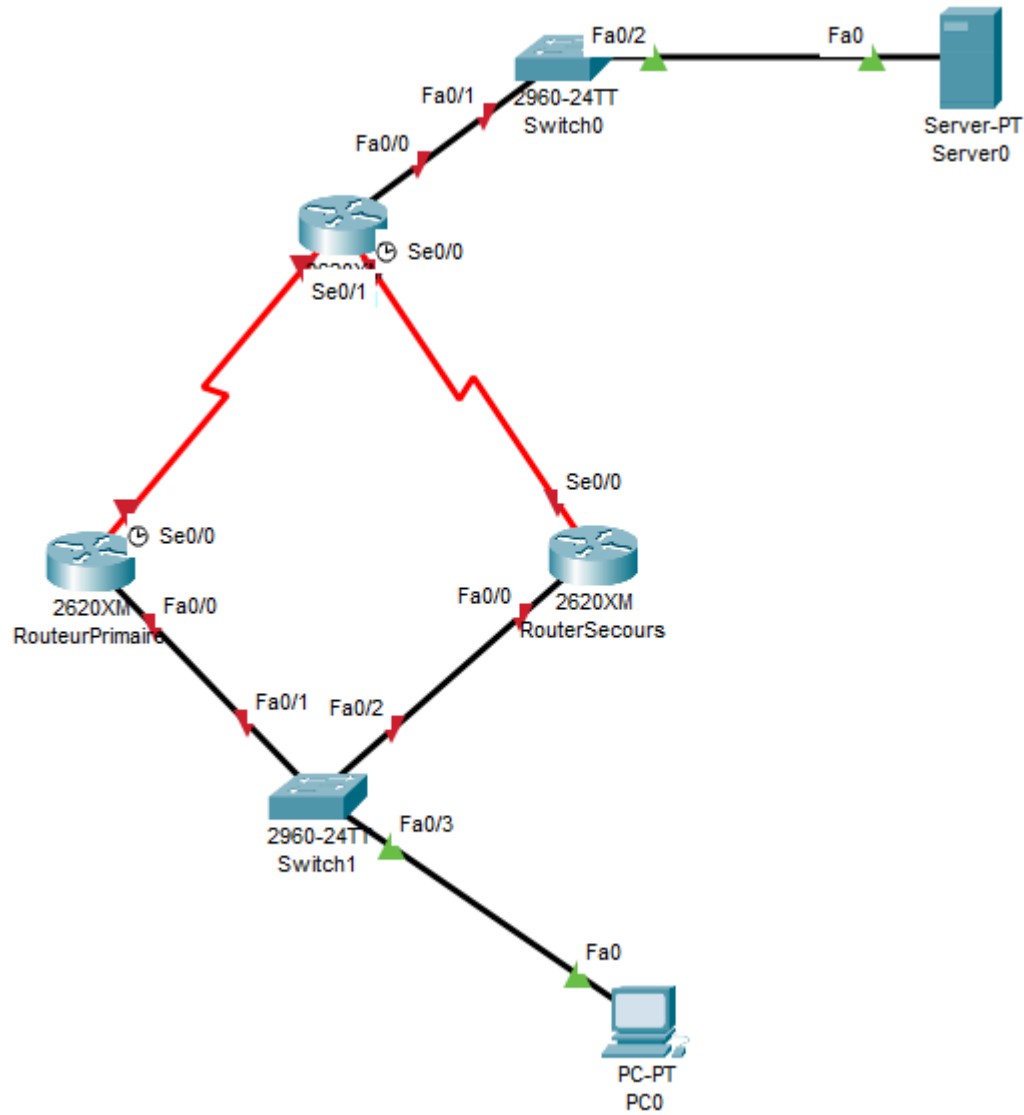
Routeurs

GABIN

Floralyn

10/10/2025

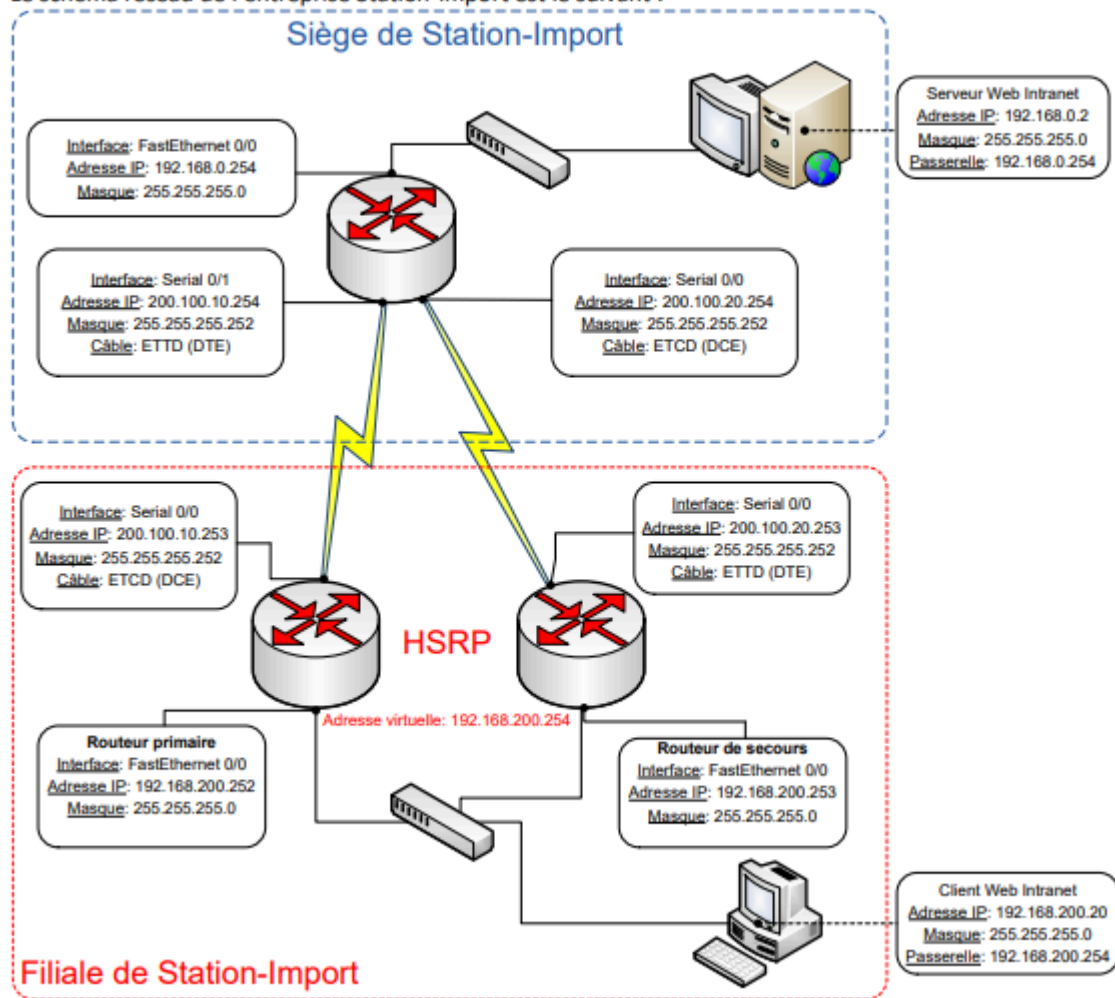
- ☐ Objectifs :
Mettre en place
la redondance
de routeurs.

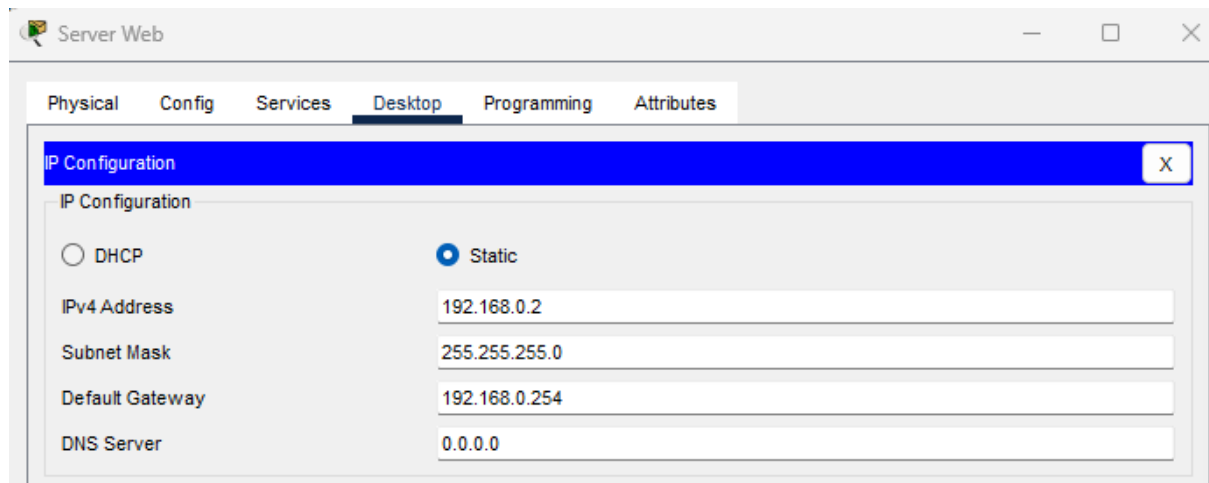


Résultat attendu à la fin du TP :

Schéma du réseau à réaliser

Le schéma réseau de l'entreprise Station-Import est le suivant :





Routeur secondaire

```

Device Name: RouterSecours
Device Model: 2620XM
Hostname: Router

Port          Link   IP Address      IPv6 Address
FastEthernet0/0 Up     192.168.200.253/24 <not set>
Serial0/0     Up     200.100.20.253/30  <not set>
Serial0/1     Down   <not set>         <not set>

Physical Location: Intercity > Home City > Corporate Office

```

Routeur du siège

```

Device Name: Router1
Device Model: 2620XM
Hostname: Router

Port          Link   IP Address      IPv6 Address
FastEthernet0/0 Up     192.168.0.254/24  <not set>
Serial0/0     Up     200.100.20.254/30 <not set>
Serial0/1     Up     200.100.10.254/30 <not set>

Physical Location: Intercity > Home City > Corporate Office

```

Étape 2 - Mettre en place le réseau

L'adresse IP du serveur : 192.168.0.2

HTTP activé :

Physical Config **Services** Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

HTTP

HTTP



On



Off

HTTPS



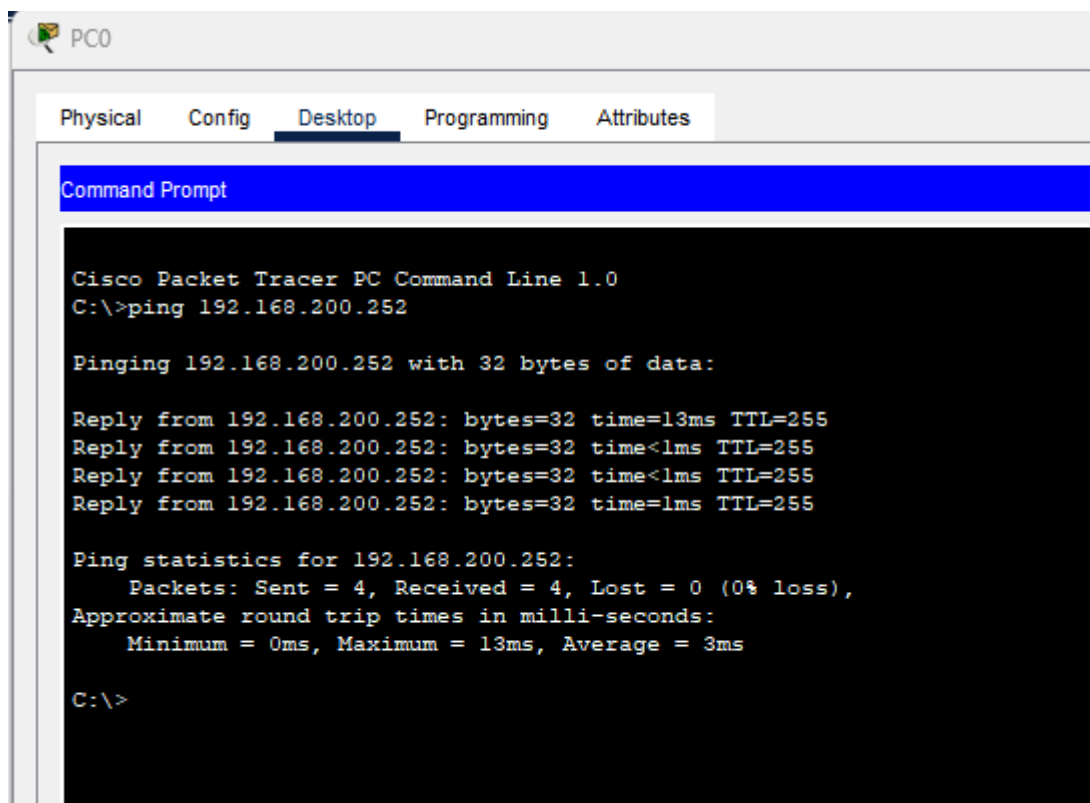
On

File Manager

	File Name	Edit	
1	copyrights.html	(edit)	
2	helloworld.html	(edit)	
3	image.html	(edit)	
4	index.html	(edit)	

Étape 4 - Vérifiez la connectivité

ping routeur primaire, secondaire OK



The screenshot shows a PC configuration window for PC0 in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of the command 'ping 192.168.200.252'. The output indicates that the ping was successful, with 4 packets sent and received, 0% loss, and round-trip times ranging from 0ms to 13ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.200.252

Pinging 192.168.200.252 with 32 bytes of data:

Reply from 192.168.200.252: bytes=32 time=13ms TTL=255
Reply from 192.168.200.252: bytes=32 time<1ms TTL=255
Reply from 192.168.200.252: bytes=32 time<1ms TTL=255
Reply from 192.168.200.252: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.200.252:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>
```

De routeur en routeur : validé

```
Router>ping 200.100.20.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.100.20.254, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/16/46 ms

Router>
```

```
Router#ping 200.100.10.254
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 200.100.10.254, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/16/43 ms
```

```
Router#
```

Étape 6 - la connectivité

→ "show ip route"
routeur secondaire :

```
Gateway of last resort is not set
```

```
R    192.168.0.0/24 [120/1] via 200.100.20.254, 00:00:27, Serial0/0
```

```
C    192.168.200.0/24 is directly connected, FastEthernet0/0
```

```
R    200.100.10.0/24 [120/1] via 200.100.20.254, 00:00:27, Serial0/0
```

```
                [120/1] via 192.168.200.252, 00:00:05, FastEthernet0/0
```

```
200.100.20.0/30 is subnetted, 1 subnets
```

```
C        200.100.20.252 is directly connected, Serial0/0
```

Étape 8 - Testez la mise en place du protocole HSRP

routeur secours :

```
show standby
```

```
FastEthernet0/0 - Group 5
```

```
State is Speak
```

```
7 state changes, last state change 01:56:47
```

```
Virtual IP address is 192.168.200.254
```

```
Active virtual MAC address is 0000.0C07.AC05
```

```
Local virtual MAC address is 0000.0C07.AC05 (v1 default)
```

```
Hello time 3 sec, hold time 10 sec
```

```
Next hello sent in 1.5084e+16 secs
```

```
Preemption enabled
```

```
Active router is 192.168.200.252
```

```
Standby router is 192.168.200.253
```

```
Priority 120 (configured 120)
```

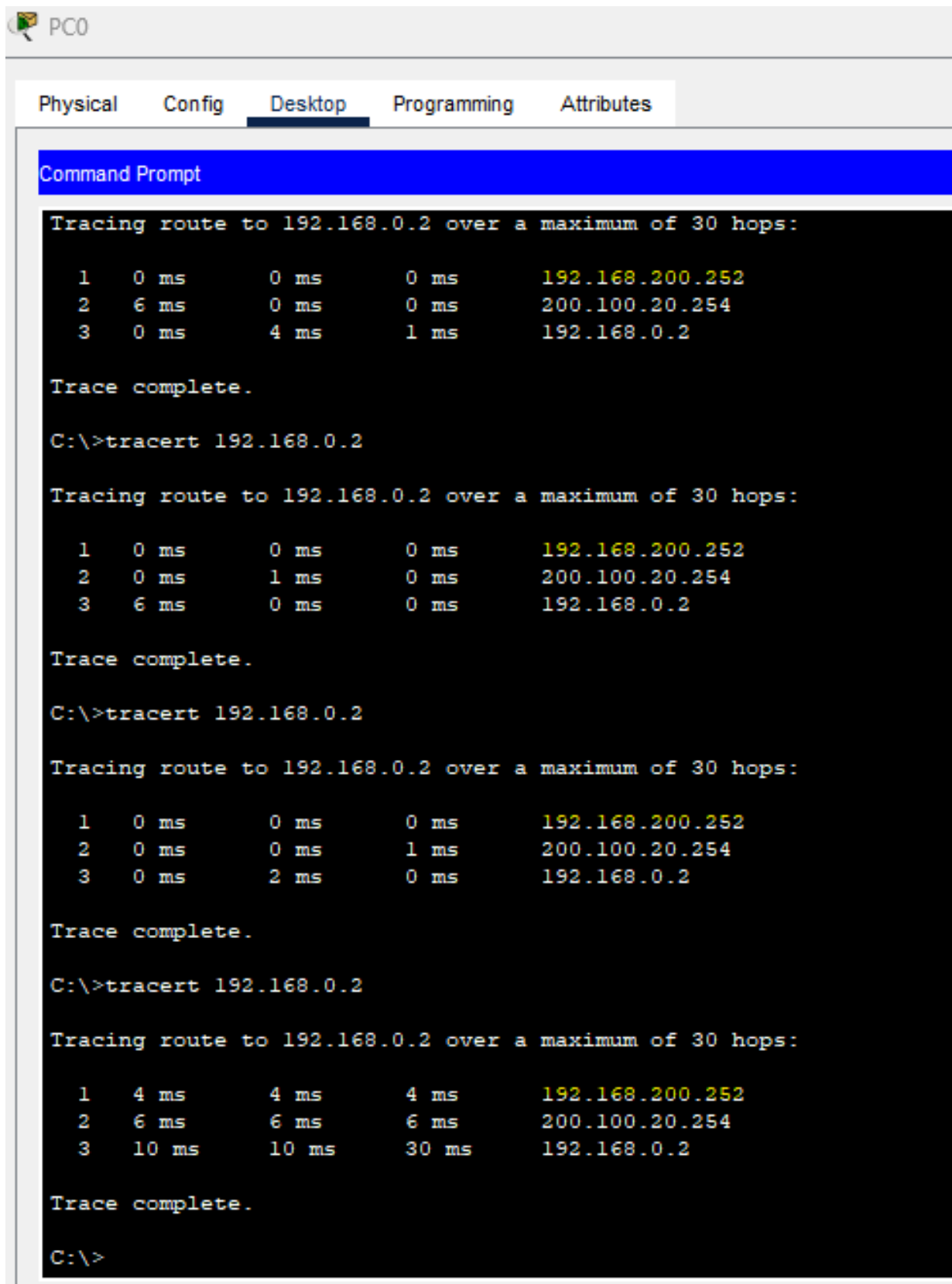
```
Track interface Serial0/0 state Up decrement 10
```

```
Group name is hsrp-Fa0/0-5 (default)
```

```
Router#
```

```
%HSRP-6-STATECHANGE: FastEthernet0/0 Grp 5 state Speak -> Standby
```

Le routeur primaire a bien été traversé :



The screenshot shows a PC0 window with a tabbed interface. The 'Desktop' tab is active, displaying a Command Prompt window. The Command Prompt shows four consecutive 'tracert' commands being executed, each tracing the route to the IP address 192.168.0.2. The results of each trace are displayed below the command, showing the path taken and the time taken at each hop. The paths are consistent across all four traces, indicating a stable network configuration.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Tracing route to 192.168.0.2 over a maximum of 30 hops:
  1  0 ms    0 ms    0 ms    192.168.200.252
  2  6 ms    0 ms    0 ms    200.100.20.254
  3  0 ms    4 ms    1 ms    192.168.0.2
Trace complete.
C:\>tracert 192.168.0.2
Tracing route to 192.168.0.2 over a maximum of 30 hops:
  1  0 ms    0 ms    0 ms    192.168.200.252
  2  0 ms    1 ms    0 ms    200.100.20.254
  3  6 ms    0 ms    0 ms    192.168.0.2
Trace complete.
C:\>tracert 192.168.0.2
Tracing route to 192.168.0.2 over a maximum of 30 hops:
  1  0 ms    0 ms    0 ms    192.168.200.252
  2  0 ms    0 ms    1 ms    200.100.20.254
  3  0 ms    2 ms    0 ms    192.168.0.2
Trace complete.
C:\>tracert 192.168.0.2
Tracing route to 192.168.0.2 over a maximum of 30 hops:
  1  4 ms    4 ms    4 ms    192.168.200.252
  2  6 ms    6 ms    6 ms    200.100.20.254
  3  10 ms   10 ms   30 ms    192.168.0.2
Trace complete.
C:\>
```

Étape 9 - Simulez une panne et testez la continuité de service

Les pings entre le pc et le serveur marchent très bien :

```
Pinging 192.168.0.2 with 32 bytes of data:
```

```
Reply from 192.168.0.2: bytes=32 time=10ms TTL=126
```

```
Reply from 192.168.0.2: bytes=32 time=2ms TTL=126
```

```
Reply from 192.168.0.2: bytes=32 time=2ms TTL=126
```

```
Reply from 192.168.0.2: bytes=32 time<1ms TTL=126
```

```
Ping statistics for 192.168.0.2:
```

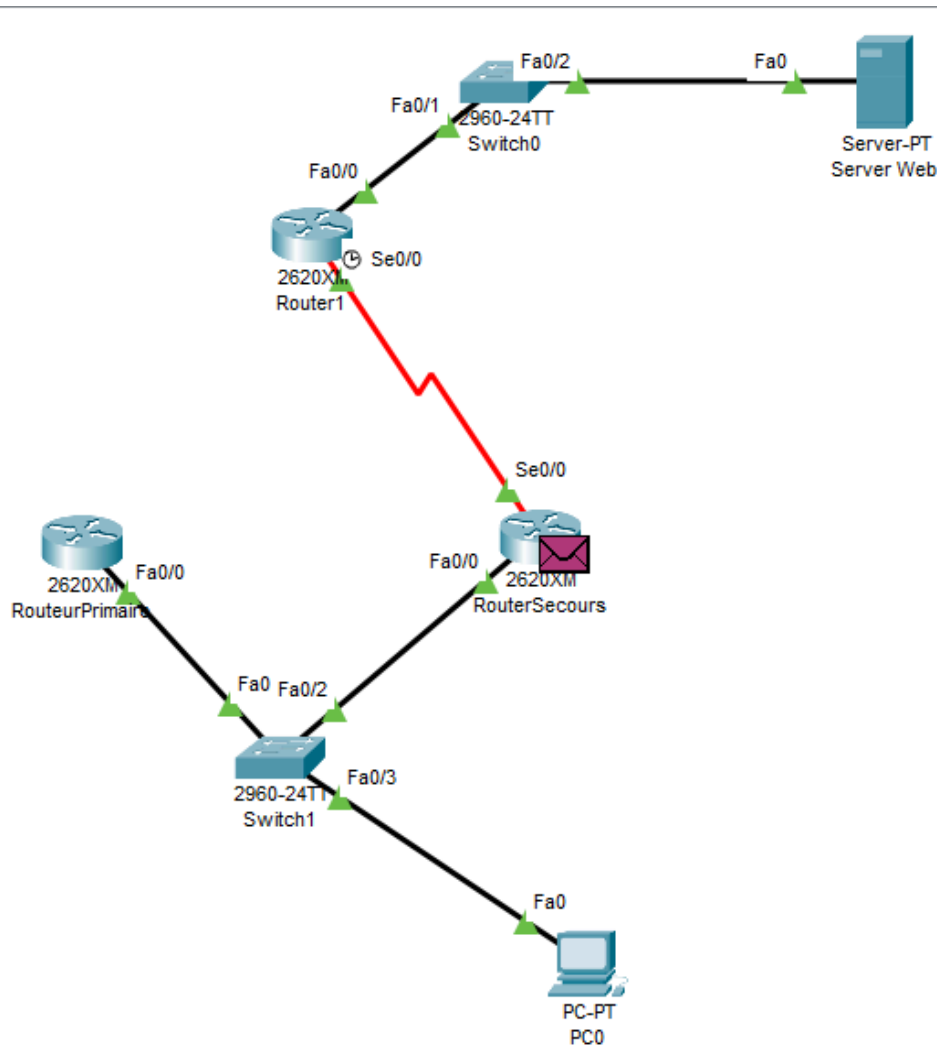
```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 10ms, Average = 3ms
```

```
C:\>
```

La requête vers le serveur web sans routeur primaire, traverse bien le routeur secondaire :



Event List		
Vis.	Time(sec)	Last Device
	0.423	Switch1
	0.424	RouterSecours
	0.424	--
	0.425	Router1
	0.426	RouterSecours
	0.427	Switch1
	0.529	--
	0.530	PC0
	0.531	Switch1
	0.532	RouterSecours
	0.532	--
	0.533	Router1
	0.534	RouterSecours
	0.535	Switch1
	0.639	--
	0.640	PC0
	0.641	Switch1
	0.642	RouterSecours
	0.643	Router1
	0.644	Switch0
	0.645	Server Web
	0.646	Switch0
	0.647	Router1

Reset Simulation ☒ Constant Delay Captured to 0.647

```
C:\>tracert 192.168.0.2
```

```
Tracing route to 192.168.0.2 over a maximum of 30 hops:
```

```

  1    4 ms    4 ms    4 ms    192.168.200.253
  2    6 ms    6 ms    6 ms    200.100.20.254
  3   10 ms   10 ms   10 ms    192.168.0.2

```

```
Trace complete.
```

La pc communique bien avec le serveur :



Étape 10 - Rétablissez la liaison et voyez la reprise sur l'activité

Voir la feuille pour plus de détails.